

Fig. 1: Lifetime vs junction temperature. Life expectancy of LEDs correlates closely with the temperature of the depletion barrier. With optimised cooling, +80°C instead of +120°C increases the life expectancy toward 100,000 hours

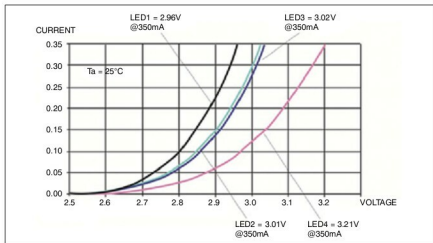


Fig. 2: Current-voltage curve of high-power LEDs

dividual LEDs, it becomes paramount that the rating graphs match while connecting a string of LEDs for lighting purposes. Connecting several LEDs in parallel to the same voltage source leads to different degrees of brightness. The human eye is very responsive to this, particularly, at low light intensity, so a solution was required.

Manufacturers could, of course, select LEDs for identical threshold voltages. But since they already select LEDs for different colour temperatures (binning), this would be an exceptionally expensive process. Some manufacturers focus on constant-current power supplies, which use the principle of

brightness of individual LEDs being proportional to the current flow, irrespective of the threshold voltage of individual diodes.

Which outlives the other—driver or LED?

Many products in daily use are 'service-free.' For example, when the batteries of an iPod reach the end of their life (about five years), it is usually time to switch to the next generation product. The same can be expected in the lighting area. Selecting the right LED driver is important for the reliability and life time of the lighting system. 50,000 hours of lighting is an extremely

long operating life—equating to seventeen years at eight hours daily. By then the lamp is amortised. However, what happens when the driver packs up after only a few years' operation?

The logistics alone can be very costly for the manufacturer, besides the loss of image. It is therefore not economic to just consider the price. Specifications of different manufacturers are similar at first glance. What differentiates these is the life expectancy. Many data sheets focus on the mean time between failure (MTBF), which provides only limited information about the quality of the product. For one, it is a calculated value that is based on individual components and refers to the time between two failures, and does not take into consideration deficiencies in manufacture.

Some manufacturers therefore specify an additional design lifetime for their new drivers which may correspond to 70,000 hours!

In-house development not really justified

Manufacturers with in-house electronics know-how and large production volume might find their own development justified. They should, however, take into account that their products must pass electromagnetic compatibility tests and meet the new UL 8750 safety standards before being certified. In many cases, it therefore proves sensible to fall back on cost-efficient solutions of certified drivers from reliable manufacturers.

The way forward

The accelerated development of LEDs drives the market for LED drivers. Converters that match the life expectancy of LEDs are in demand. Service costs tend towards zero provided matching life expectancy is guaranteed. Wrong cost savings might prove expensive. High warranty or a specified life expectancy might be helpful to eliminate uncertainties, especially from reliable manufacturers who will be active in the market for years to come. ●

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